

Samrudha Narendra

Bengaluru, India

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EDUCATION

PES University

B.Tech in Computer Science Engineering — CGPA: 8.25/10

2023 – 2027

Bengaluru, India

National Public School, Rajajinagar

Senior High School — Percentage: 91.2%

2019 – 2023

EXPERIENCE

Summer Research Intern

CCBD-CDASML Research Centre, PES University

June 2025 – July 2025

Bengaluru, India

- Fine-tuned OpenAI Whisper ASR models on Kannada speech corpora with phoneme-level annotations using PyTorch, targeting WER/CER reduction across gender-stratified evaluation sets
- Built automated evaluation pipelines computing WER and CER metrics across gender-specific datasets, surfacing key accuracy gaps and alignment constraints affecting model performance
- Analyzed dataset variability and audio alignment constraints, producing findings that directly informed model retraining strategy
- Published peer-reviewed research at IEEE I3CTCON 2026 [\[Paper\]](#)

Teaching Assistant – C Programming

PES University

Feb 2026 – Present

Bengaluru, India

- Built an agentic RAG pipeline with hybrid BM25 + FAISS retrieval, automated hallucination detection, query reformulation, and self-correction workflows; validated on HotpotQA benchmarks with iterative re-retrieval loops

PROJECTS

Hybrid AI Product Search Engine | *Python, FastAPI, FAISS, BM25, LightGBM, Kafka, MongoDB, React, Docker*

- Architected a production-grade multi-stage retrieval pipeline combining BM25 lexical search, FAISS semantic retrieval, and LightGBM LambdaRank re-ranking, achieving sub-second search latency at scale
- Implemented Reciprocal Rank Fusion (RRF) with ML-based re-ranking and real-time filtering on category, price, and inventory constraints, significantly improving result relevance and ranking quality
- Designed a decoupled microservices architecture using FastAPI, Kafka, MongoDB, and Docker, supporting real-time product indexing, live updates, and horizontal scaling

Self-Healing Agentic RAG System | *Python, LangChain, FAISS, BM25*

- Built a Retrieval-Augmented Generation pipeline with hybrid BM25 + FAISS retrieval, automated hallucination detection, and iterative query reformulation for multi-hop reasoning tasks
- Implemented self-correction workflows with re-retrieval loops on HotpotQA benchmarks, boosting answer faithfulness and reducing hallucination rate through iterative self-healing

RideSync – Secure Campus Ride-Sharing Platform | *Java, Spring Boot, JWT, Spring Security, JavaFX, MySQL*

- Delivered a full-stack campus ride-sharing platform with role-based access control for riders, drivers, and admins; secured 40+ REST APIs using JWT authentication and Spring Security
- Implemented route optimization using Dijkstra's algorithm with OTP verification, banned-user enforcement, ride history, ratings, and audit logging in a layered MVC architecture with JPA repositories

TECHNICAL SKILLS

Languages: Python, Java, C, JavaScript, SQL

Frameworks & Libraries: React.js, Node.js, Express.js, Spring Boot, FastAPI, PyTorch, LangChain

Machine Learning / AI: NLP, Transformers, Whisper ASR, RAG, FAISS, BM25, LambdaRank, Hallucination Detection

Databases: MongoDB, MySQL

Tools & Technologies: Git, GitHub, Docker, Kafka, Postman, Linux, REST APIs

Core CS Subjects: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP

Club Head – GronIT Club, PES University

2025 – 2026

- Spearheaded ideathons and a state-level hackathon with 100+ participants; managed sponsorships, outreach, logistics, and a 60-member organizing team end-to-end
- Drove club growth through strategic partnerships and cross-campus outreach, expanding event reach and participant engagement year-over-year

Event Management Core – Aikya Club, PES University

2025 – 2026

- Planned and executed 5+ community and technical events; coordinated cross-functional teams of 10–20 members across logistics, venue, and operations